Non-Newtonian Viscosity

Final Lab Report
Unit Operations Lab 2
September 29, 2025

Group 3
Lindsey Meisch
Basil Hashim
Kendall Johnson

1. Executive Summary (10 pts)

A capillary tube viscometer was used to determine the viscous properties of the Non-Newtonian Fluid made of 1wt% CARBOPOL 934 at pH 6. The capillary tube viscometer was chosen as it is simple and consistent. For the data collection, the mass of fluid that flows through a specific size of capillary tube under pressure during a specific amount of time was used to determine the mass flow rate; which when divided by the collected density yielded the volumetric flow rate. A total of six different capillary tubes were used within this experiment, and each capillary tube was evaluated at three different pressures. Capillary tubes had either one of two lengths, 6 or 9inches, with one of three diameters, 0.053, 0.062, or 0.070inches. Power-law constants, m and n, were solved for within this lab, under the given information. Using the relationship (University of Illinois Chicago, 2023),

$$\frac{\Delta P\, R}{2L} = m\left[\frac{3n+1}{n}\cdot\frac{Q}{\pi R^3}\right]^n$$

the power-law constants could be solved for. It was determined within this lab that the solution was a shear thinning material; as the n-values solved for were less than 1 (range: 0.45 – 0.66). Values of n solved for from the 9inch capillary tubes were consistent with those anticipated in literature for such a solution (range: 0.45-0.48). The fluid exhibited pseudoplastic stress-strain behavior.

2. Introduction (10 pts)

The objective of this experiment was to measure the viscosity of a Non-Newtonian Fluid, in this case, CARBOPOL 934, using a capillary tube viscometer. Viscosity is a key transport property that has relativity to momentum, heat and mass transfer. Its accurate determination is essential for designing and optimizing operations such as pumping, mixing and heat exchange (Unit Operations Lab Manual, 2023). Non-Newtonian fluids such as polymer dispersions, gels and food suspension exhibit shear-dependent viscosities that require more sophisticated methods of characterization than Newtonian fluids (Al-Malah, 2006).

The capillary tube viscometer is widely used due to its simplicity, accuracy and close resemblance to industrial flow conditions such as extrusion and pipeline transport (Shin & Keum, 2003). Early designs of this equipment such as the modified U-tube introduced in the mid-20th century, emphasized reducing experimental error from kinetic energy corrections and facilitating rapid measurements (Wagner & Russell, 1948). More recently, new innovations such as mass-detecting capillary viscometers, extend the measurable shear rate range and improve efficiency for non-Newtonian systems (Shin & Keum, 2003).

The significance of this experiment was in the understanding of the rheological behavior of CARBOPOL dispersions. Carbomers are cross-linked polyacrylic acid polymers that act as highly efficient thickeners and stabilizers in a wide range of pharmaceutical, cosmetic and food products (Ashland, 2011). The viscosity of these products depends on the concentration, pH and neutralization conditions (Lubrizol, 2011). By measuring and monitoring their Non-Newtonian flow behavior, one can better understand their relevance in industries ranging from personal care to pharmaceuticals to coatings and food engineering.

The apparatus that was used in this experiment consists of a pressurized vessel connected to interchangeable precision-bore capillary tubes. When a known pressure drop is applied across the tube and measuring the corresponding volumetric flow rate, the viscosity of the CARBOPOL solution can be determined. Experimental data was analyzed using rheological models to extract flow parameters and assess deviations from Newtonian behavior.

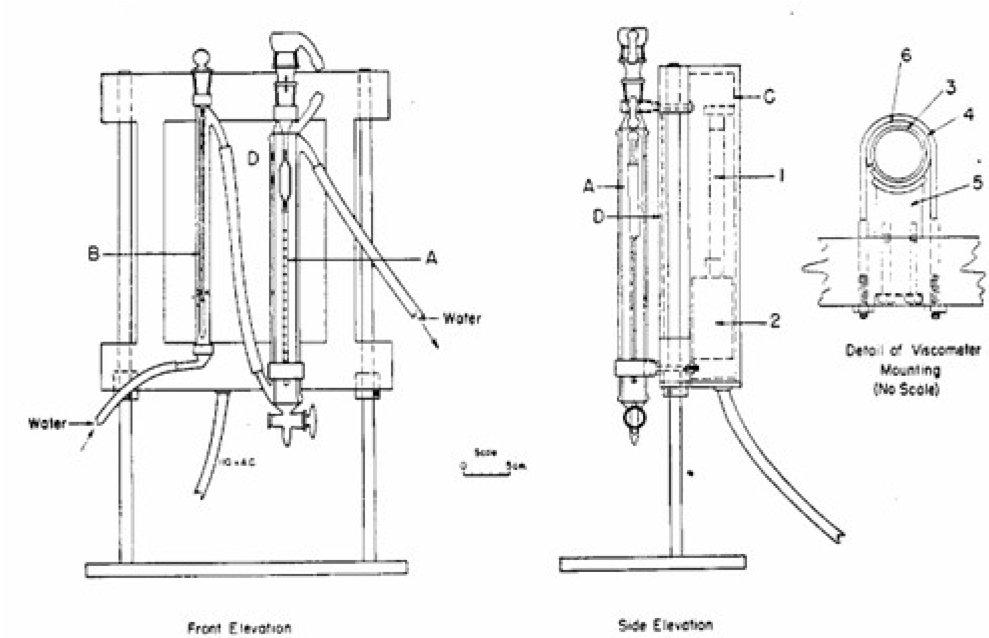


Figure 1. Capillary-tube viscometer design used for polymer solutions. Wagner & Russell (1948).

3. Theory (10 pts)

Viscosity can be defined as the ratio of shear stress to shear rate in a fluid. For Newtonian fluids, the relationship is linear and constant at all shear rates. This is expressed by Newton's law of viscosity (University of Illinois Chicago, 2023):

$$\tau = \mu \frac{dv}{dy}$$

Where τ is the shear stress, μ the viscosity and dv/dy the velocity gradient. Non-Newtonian fluids deviate from this linearity. Their relationship between shear stress and shear rate can be described by the Ostwald-de Waele (power law) model (University of Illinois Chicago, 2023):

$$\tau = m \left| \frac{dv}{dy} \right|^n$$

Where *m* is the consistency index and n is the flow behavior index. For $n < 1$, the fluid is shear-thinning and known as a pseudoplastic. For $n > 1$, it is shear-thickening, known as dilatant. CARBOPOL dispersions above ~0.45 wt% typically exhibit shear-thinning behavior, transitioning from Newtonian response at lower concentrations (Al-Malah, 2006).

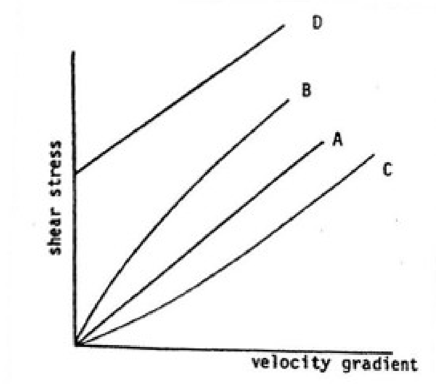


Figure 2. Comparison of Newtonian and Non-Newtonian shear stress-shear rate relationships. University of Illinois Chicago (2023).

The viscosity of a Newtonian fluid in a capillary tube viscometer can be determined from the Hagen-Poiseuille equation (Son, 2007):

$$Q = \frac{\Delta P\,\pi R^4}{8\mu L}$$

Where *Q* is the volumetric flow rate, *ΔP* is the pressure drop, *R* is the tube radius and L is the tube length. The relation for non-Newtonian power-law fluids is modified to (University of Illinois Chicago, 2023):

$$\frac{\Delta P\,R}{2L} = m\left[\frac{3n+1}{n}\cdot\frac{Q}{\pi R^3}\right]^n$$

From experimental measurements of *ΔP* and *Q*, the power-law constants *m* and *n* can be determined. A log-log plot of shear stress vs shear rate yields a slope equal to *n* and an intercept related to *m* (University of Illinois Chicago, 2023).

CARBOPOL rheology is highly sensitive to neutralization and pH; when neutralized to a pH between 6 and 8, a cross-linked network is formed in CARBOPOL that dramatically increases viscosity which can often exceed 50,000 cP at 1 wt% concentration (Lubrizol, 2011; Ashland, 2011). Changes in polymer concentration or the addition of electrolytes can further alter the flow behavior which can make the experimental characterization essential for predictive modeling (Ashland, 2011).

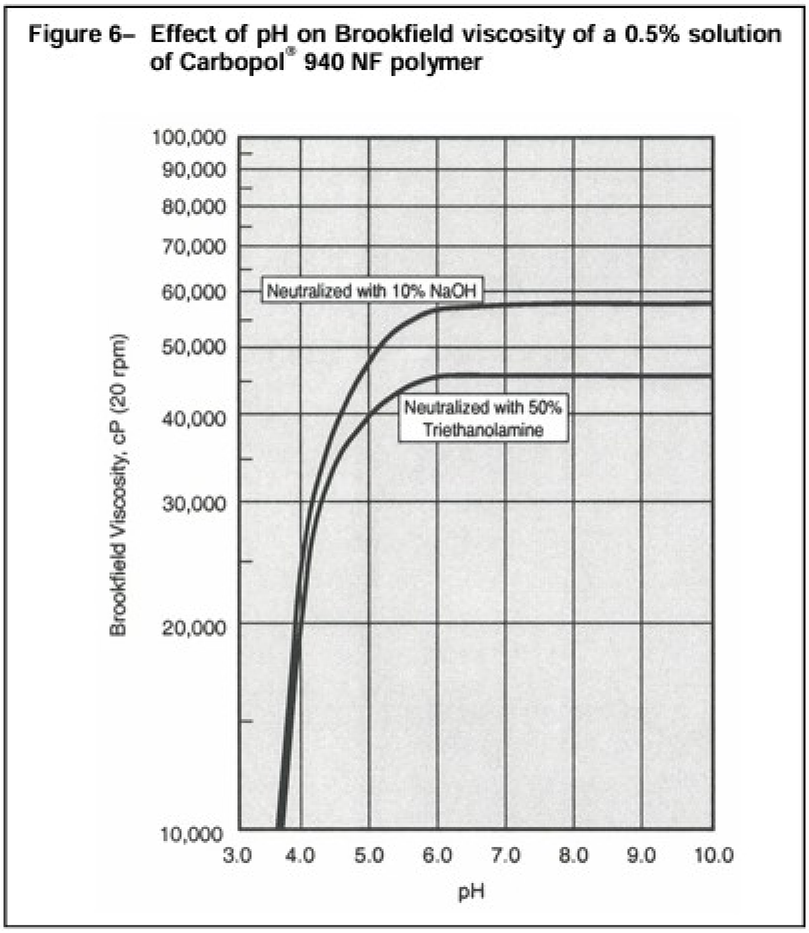
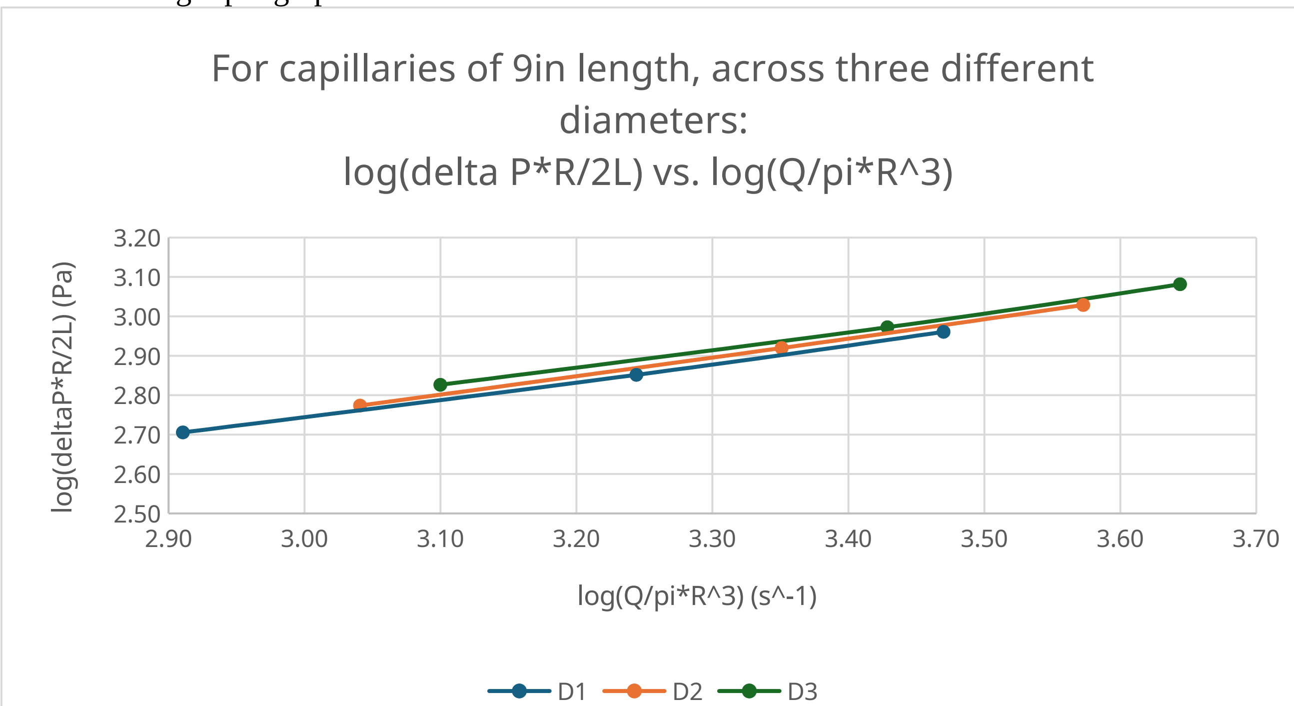


Figure 3. Effect of pH on viscosity of a 0.5% CARBOPOL 940 NF aqueous dispersion. Lubrizol Advanced Materials, Inc. (2011).

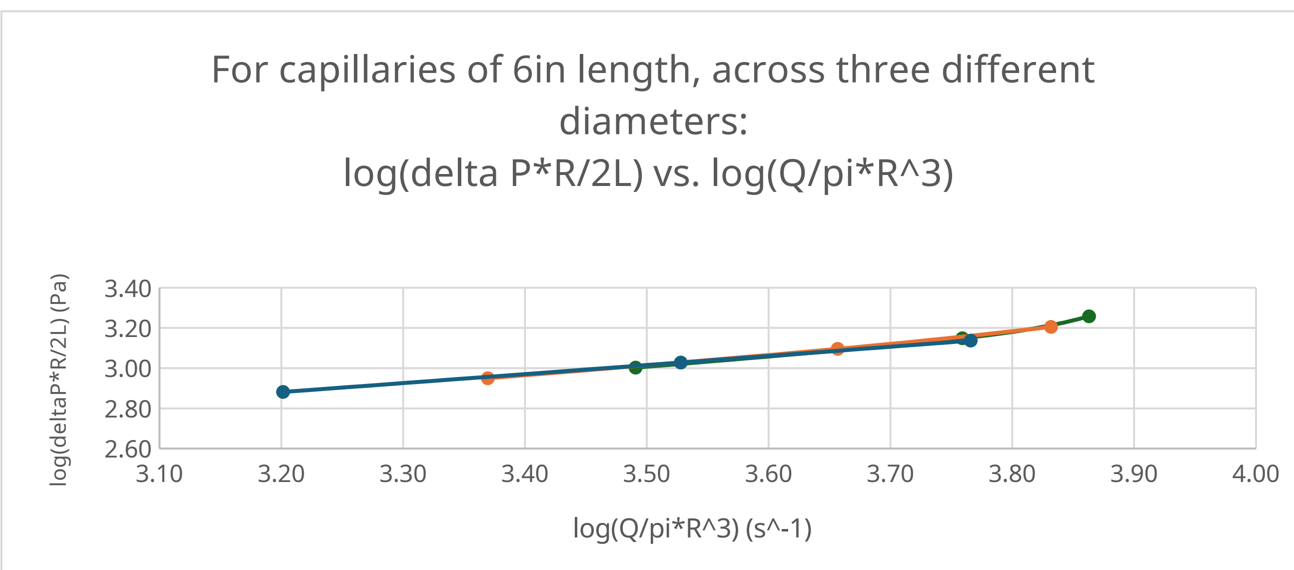
In this experiment, the collected data of the flow rates vs pressure was analyzed to determine whether the prepared CARBOPOL solution behaves as a Newtonian or Non-Newtonian fluid and rheological constants were extracted accordingly. The results were then compared against published correlations for CARBOPOL.

4. Results (15 pts)

The following two graphs were made to show the relationship that generated respective m and n values, and show the three different diameters performances across one consistent length per graph.



Graph 1: For three capillary tubes of 9in length, with D1 being 0.053in, D2 being 0.062in, and D3 being 0.07in, a log/log plot to solve for both m and n values.



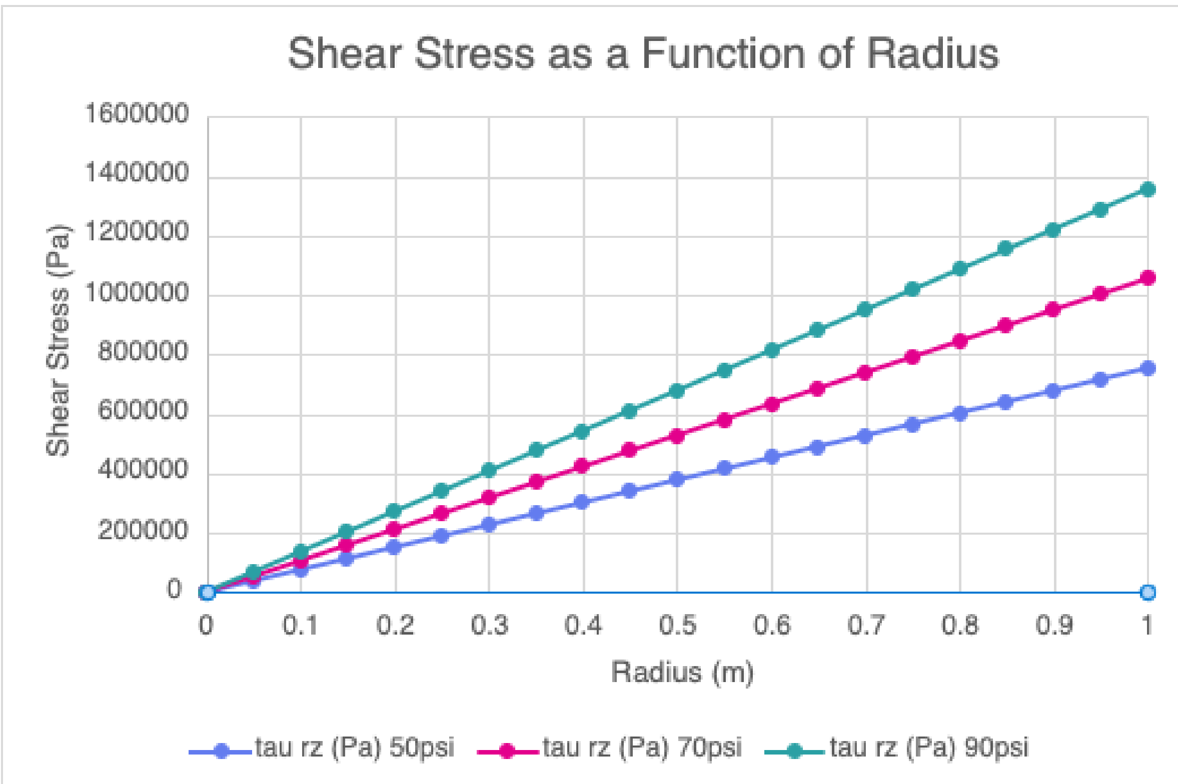
Graph 2: For three capillary tubes of 6in length, with D1 being 0.053in, D2 being 0.062in, and D3 being 0.07in, a log/log plot to solve for both m and n values.

The following table lists the individual m and n values for each capillary condition, as well as the L/D values for each condition.

	Length: 6in			Length: 9in		
	m (Pa*s)	n (Pa)	L/D	m (Pa*s)	n (Pa)	L/D
Diameter: 0.053in	1.99	0.45	113.21	1.88	0.45	169.81
Diameter: 0.062in	1.27	0.55	96.77	1.71	0.48	145.16
Diameter: 0.070in	0.75	0.66	85.71	1.84	0.47	128.57

Table 1: Respective m, n, and L/D values for all six capillary tubes tested within this experiment.

The following graph is a representation of the shear stress Tau rz as a function of radius for the capillary tubes of 9inches.



Graph 3: A graphical representation of the shear stress Tau rz, as a function of radius for capillary tubes of length 9inches.

5. Discussion (15 pts)

The main values calculated in this lab were the m and n values for the flowrate of 1 wt% Carbopol solution equation. The results in **Table 1** (n-values: 0.45 – 0.48) match the n-values reported in *Rheological Characterizations of Carbopol® Dispersions in Water and in Water/Glycerol Solutions*, where n values between 0.384 and 0.480 were reported when solutions between 1 and 3 wt% were tested using a rheometer.

The n value is used to assess the nature of a non-Newtonian fluid. An $n < 1$ means that the Carbopol solution is a shear-thinning material; a material with a viscosity that decreases as increasing shear is applied. This relation also matches the data shown in **Graph 3**, further corroborating the experimental findings**.** This dataset also shows that significant pseudoplastic behavior occurs as the Carbopol solution moves through the capillary tubes, as seen by the non-linear shear vs. stress relationship that was plotted. The relationship seen correlates to the shear-thinning (viscoelastic) curve shown in the literature:

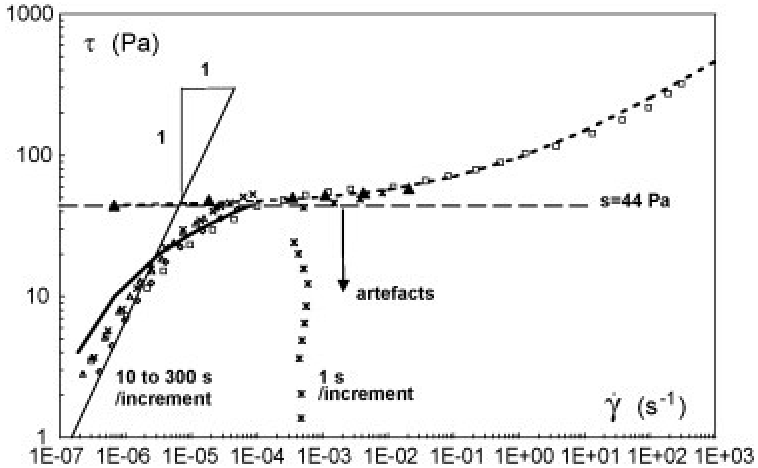


Figure 4. Stress vs. Strain curve of neutralized 0.5 wt% Carbopol Solution. [2]

Although the experimental values generally agree with the literature and can be used to definitively conclude that the 1 wt% Carbopol solution is shear-thinning and that it exhibits pseudoplastic stress-strain behavior, several experimental constraints prevent more specific or more quantitative claims from being formed. A major source of potential error stems from the experimental procedure for measuring the mass flow of the solution. Human error was present in the opening/closing of valves and timing of the flow. A slight pressure drop was also observed at the start of each trial, although the magnitude of this drop was extremely limited. Additionally, it was assumed that the viscometer operated at steady state, however it took a potentially non-negligible amount of time for the device to reach this state once an experimental trial was started. However, the overall agreement between the sources and the experimental data demonstrates that neglecting the pressure drop and the transient flow are valid assumptions.

If the experiment were to be performed again, completing multiple trials at each data point could potentially mitigate the error caused by the above reasons. A larger range of capillary tube diameters and lengths may also provide greater insight into the fluid's behavior. Completing the experiment with lower flowrates (and consequentially lower shear rates) may also yield results more comparable to those cited, however this is difficult to do with the available apparatus. Compared to many of the slit rheometers used within the literature, the capillary tube viscometer is far less sensitive due to its reliance on a pressure drop to produce a flow through the device, limiting its utility in low-flowrate instances as well as the scope of the experimental observations presented above. Another potential source of deviation from literature values is the pH and composition of the Carbopol solution used in the experimental trials, as this determines the crosslinking—and resulting properties—of the gel, so slight variations in pH may have affected the fit of the data collected. The solution used in the experiment had a final pH of 6.

6. References (10 pts)

[1] R. Varges, Priscilla, Camila M. Costa, Bruno S. Fonseca, Mônica F. Naccache, and Paulo R. De Souza Mendes. "Rheological Characterization of Carbopol® Dispersions in Water and in Water/Glycerol Solutions." MDPI, January 4, 2019. https://www.mdpi.com/2311-5521/4/1/3.

[2] Piau, J. M. Carbopol Gels: Elastoviscoplastic and Slippery Glasses Made of Individual Swollen Sponges: Meso- and Macroscopic Properties, Constitutive Equations and Scaling Laws. 2007, 144 (1), 1–29. https://doi.org/10.1016/j.jnnfm.2007.02.011.

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[4] Ashland. (2011). *Ashland™ 940 and 980 carbomers: Essential rheology modifiers for personal care formulating*. Ashland Inc.

[5] Lubrizol Advanced Materials, Inc. (2011, May 31). *Pharmaceutical bulletin 6: Thickening properties* (Previous editions: 2004, 2008). Lubrizol Corporation.

[6] Shin, S., & Keum, D.-Y. (2003). Viscosity measurement of non-Newtonian fluid foods with a mass-detecting capillary viscometer. *Journal of Food Engineering, 58*(1), 5–10. https://doi.org/10.1016/S0260-8774(02)00327-8

[7] Son, Y. (2007). Determination of shear viscosity and shear rate from pressure drop and flow rate relationship in a rectangular channel. *Polymer*, *48*(2), 632–637. https://doi.org/10.1016/j.polymer.2006.11.048

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[9] Wagner, R. H., & Russell, J. (1948). Capillary-tube viscometer for routine measurements of dilute high-polymer solutions. *Analytical Chemistry, 20*(2), 151–155. https://doi.org/10.1021/ac60016a003

7. Appendix I: Data Tabulation/Graphs (10 pts)

Begin your Data Tabulation/Graphs section here. This section will contain your raw data and measurements plus any calculated data. Tables of data can be used, but make sure they are clear and easy to read. Use an appropriate number of significant figures in your data tabulation or points will be taken off.

The fluid density was calculated by averaging the three mass values collected for one-hundred milliliters of the Carbopol solution, as done below;

$$\rho = \frac{109.7\,g + 109.2\,g + 109.7\,g}{3 * 100\,mL} = 1.10\,g/mL$$

The mass values collected with respect to different capillary tubes, under different times, and under different pressures are expressed in tables one through six:

Table 2: Capillary tube data collection of 9inch length, and 0.053inch diameter.

Table 3: Capillary tube data collection of 9inch length, and 0.062inch diameter.

Table 4: Capillary tube data collection of 9inch length, and 0.07inch diameter.

Table 5: Capillary tube data collection of 6inch length, and 0.053inch diameter.

Table 6: Capillary tube data collection of 6inch length, and 0.062inch diameter.

Table 7: Capillary tube data collection of 6inch length, and 0.07inch diameter.

Where the mass flow rate was calculated using the formula,

$$\frac{m(g)}{t(s)} = Mass\ Flow\ Rate\left(\frac{g}{s}\right)$$

And the mass flow rate was converted to a volumetric flow rate using the density,

$$\frac{Mass\ Flow\ Rate\left(\frac{g}{s}\right)}{\rho\left(\frac{g}{mL}\right)} = Volumetric\ Flow\ Rate\left(\frac{mL}{s}\right) = \left(\frac{cm^3}{s}\right)$$

Volumetric flow rate was then converted from cm^3/s to m^3/s by dividing by 100^3. This value is now Q.

Psi was converted to Pa, and length was converted to m. The radius was solved for by dividing the diameter by a factor of two.

The relationship given in the lab manual can now be used to solve for m and n, as it is now able to be filled in with the presented variables:

$$\log\left(\frac{\Delta P * R}{2L}\right) vs.\log\left(\frac{Q}{\pi * R^3}\right)$$

Where n is the slope found with this relationship and m can be solved for with the following expression:

$$\log\left\{m\left[\frac{3n+1}{n}\text{¿¿¿}n\right\}=y-intercept$$

From there, the following relationship was used to express how the shear stress changes as a function of

	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
3	Length tube (in)	6		Density (g/cm^3)	1.1										
4	Length tube (m)	0.15													
5															
6											Q				
7		Diameter (in)	Diameter (m)	Radius (m)	Pressure (psi)	Pressue (Pa)	Time (s)	Mass (g)	Mass Flow Rate (g/s)	VFR (cm^3/s)	VFR (m^3/s)	x	y	logx	logy
8		0.053	0.00135	0.000673	50	344750	15	25.14	1.68	1.52	1.52E-06	1.59E+03	761.32	3.2	2.88
9		0.053	0.00135	0.000673	70	482650	15	53.31	3.55	3.23	3.23E-06	3.37E+03	1065.85	3.53	3.03
10		0.053	0.00135	0.000673	90	620550	15	92.2	6.15	5.59	5.59E-06	5.83E+03	1370.38	3.77	3.14
11		0.062	0.00157	0.000787	50	344750	10	39.51	3.95	3.59	3.59E-06	2.34E+03	890.6	3.37	2.95
12		0.062	0.00157	0.000787	70	482650	10	76.53	7.65	6.96	6.96E-06	4.54E+03	1246.85	3.66	3.1
13		0.062	0.00157	0.000787	90	620550	10	114.53	11.45	10.41	1.04E-05	6.79E+03	1603.09	3.83	3.2
14		0.07	0.00178	0.000889	50	344750	10	75.18	7.52	6.83	6.83E-06	3.10E+03	1005.52	3.49	3
15		0.07	0.00178	0.000889	70	482650	10	139.4	13.94	12.67	1.27E-05	5.74E+03	1407.73	3.76	3.15
16		0.07	0.00178	0.000889	90	620550	5	88.56	17.71	16.1	1.61E-05	7.29E+03	1809.94	3.86	3.26
17															
18			Dimater 1		Diameter 2		Diameter 3								
19		psi	logx	logy	logx	logy	logx	logy							
20		50	3.2	2.88	3.37	2.95	3.49	3							
21		70	3.53	3.03	3.66	3.1	3.76	3.15							
22		90	3.77	3.14	3.83	3.2	3.86	3.26							
23															
24		L/D	n	int	int^e	((3n+1)/n)^n	m								
25		113.21	0.45	1.43	4.2	2.11	1.99								
26		96.77	0.55	1.1	3	2.37	1.27								
27		85.71	0.66	0.7	2.01	2.7	0.75								
28															
29		Average n	0.55			Average m	1.33								
30															
31															
32															
33			LINEST 1			LINEST 2			LINEST 3						
34			0.452036762	1.43395173		0.54801635	1.10001941		0.657859354	0.699286938					

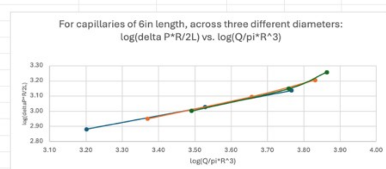

Figure 1: A screenshot of the excel calculations that were done for capillary tubes of length 6in.

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
3	Length tube (in)	9		Density (g/	1.1										
4	Length tube (m)	0.2286													
5															
6											Q				
7		Diameter (	Diameter (	Radius (m)	Pressure (p	Pressue (Pa	Time (s)	Mass (g)	Mass Flow	VFR (cm^3	VFR (m^3/	x	y	logx	logy
8		0.053	0.00135	0.000673	50	344750	30	25.73	0.86	0.78	7.80E-07	8.14E+02	507.55	2.91	2.71
9		0.053	0.00135	0.000673	70	482650	15	27.73	1.85	1.68	1.68E-06	1.75E+03	710.57	3.24	2.85
10		0.053	0.00135	0.000673	90	620550	15	46.65	3.11	2.83	2.83E-06	2.95E+03	913.59	3.47	2.96
11		0.062	0.00157	0.000787	50	344750	15	27.8	1.85	1.68	1.68E-06	1.10E+03	593.74	3.04	2.77
12		0.062	0.00157	0.000787	70	482650	15	56.77	3.78	3.44	3.44E-06	2.24E+03	831.23	3.35	2.92
13		0.062	0.00157	0.000787	90	620550	15	94.61	6.31	5.73	5.73E-06	3.74E+03	1068.73	3.57	3.03
14		0.07	0.00178	0.000889	50	344750	10	30.56	3.06	2.78	2.78E-06	1.26E+03	670.35	3.1	2.83
15		0.07	0.00178	0.000889	70	482650	10	65.15	6.52	5.92	5.92E-06	2.68E+03	938.49	3.43	2.97
16		0.07	0.00178	0.000889	90	620550	10	106.95	10.7	9.72	9.72E-06	4.40E+03	1206.63	3.64	3.08
17															
18			Diameter 1		Diameter 2		Diameter 3								
19		psi	logx	logy	logx	logy	logx	logy							
20		50	2.91	2.71	3.04	2.77	3.1	2.83							
21		70	3.24	2.85	3.35	2.92	3.43	2.97							
22		90	3.47	2.96	3.57	3.03	3.64	3.08							
23															
24		L/D	n	int	int^e	((3n+1)/n)^	m								
25		169.81	0.45	1.38	3.97	2.12	1.88								
26		145.16	0.48	1.32	3.73	2.18	1.71								
27		128.57	0.47	1.38	3.96	2.15	1.84								
28															
29		Average n	0.47			Average m	1.81								
30															
31															
32															
33			LINEST 1			LINEST 2			LINEST 3						
34			0.454925	1.37979		0.479379	1.315147		0.467177	1.375981					

Figure 2: A screenshot of the excel calculations that were done for capillary tubes of length 9in.

8. Appendix II: Error Analysis (10 pts)

The density is 1.10 g/mL +/- 0.003 g/mL.

No other repeated measurements or averages were taken within this lab.

9. Appendix III: Sample Calculations (10 pts)

I used the capillary tube with length 9inches, and 0.053inch diameter, under the pressure of 50psi.

OTHER | PLANT | AFE # | PREPARED BY | SHEET ___ OF ___

Length tube (in) = 9
Diameter tube (in) = 0.053
Pressure (psi) = 50
Time (sec) = 30
Mass (g) = 25.73

$\}$ given data

$\rho = 1.1\ (g/mL) \rightarrow$ Calc. done in App. 1

Mass flow $= \dfrac{25.73g}{30sec} = 0.86\ g/s$

Vol. flow $= \dfrac{0.86g}{s} \times \dfrac{mL}{1.1g} = 0.78\ mL/s = 0.78\ cm^3/s$

Vol. flow $\dfrac{cm^3}{s} \rightarrow \dfrac{m^3}{s} = 7.80 \times 10^{-7}\ m^3/s = Q$

Length tube (m) = 0.23
Diameter tube (m) = 0.0014
Pressure (Pa) = 344750

$$X = \frac{Q}{pi \cdot R^3} = \frac{7.80 \times 10^{-7}\ m^3/s}{(\pi)\left(\frac{0.0014m}{2}\right)^3} = 814\ s^{-1}$$

$$y = \frac{\Delta P R}{2L} = \frac{(344750 Pa)\left(\frac{0.0014m}{2}\right)}{2(0.23m)} = 507.55\ Pa$$

$\log(X) = 2.91 \quad \log(y) = 2.71 \rightarrow$ Coordinates to be plotted to solve for m+n

=LINEST(yvals, xvals, true, true) was used to find m+n given log(x) log(y) across the 3 different pressures.

$\rightarrow$ slope $= n = 0.45$

$\rightarrow$ y-int $= 1.43 \rightarrow e^{1.43} = \left(\dfrac{3n+1}{n}\right)^n \rightarrow m = 1.99$

For the following shear stress calculations, I used a capillary tube of 9inches at 50psi.

Using information from the provided lab manual, App 1

$$\tau_{rz} = m\left|\frac{dV_z}{dr}\right|^n \quad \text{where} \quad \frac{dV_z}{dr} = \left(\frac{1}{m}\frac{\Delta P}{2L}\right)^{1/n} r^{1/n}$$

$$\tau_{rz} = m\left[\left(\frac{1}{m}\frac{\Delta P}{2L}\right)^{1/n} r^{1/n}\right]^n \rightarrow \tau_{rz} = \cancel{m}\frac{1}{\cancel{m}}\frac{\Delta P}{2L} r \rightarrow \tau_{rz} = \frac{\Delta P}{2L} r$$

For capillary tubes of 9inches, at 50psi

$$\tau_{rz} = \frac{(344750\ Pa)}{2(0.2286m)} r$$

where r is going to be used to show how the shear stress changes across the radius.

10. Appendix IV: Individual Team Contributions

NAME: Kendall Johnson

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Prepared the gel solution during the pre-lab, operated the viscometer in lab.
Pre-Lab Editing	1	Adjusted information as needed.
Apparatus (prelab)	1	Took a picture of the apparatus and labeled it.
Materials & Supplies (prelab)	1	Listed the materials and supplies used for the lab.
Project Safety Evaluation (prelab)	1	Evaluated different safety concerns, both during the pre-lab day in making the gel, and when operating the viscometer.
Executive Summary (final lab)	1	Summarized the lab
Results (final lab)	12	Figured out the calculations needed for the lab, created the excel file, and spent a lot of time trying to figure out the best way to display the data generated from this lab.
Data Tabulation/Graphs (final)	4	Made graphs from my excel calculations, excel ended up crashing and I had to use a different version.
Error Analysis (final lab)	1	Considered standard deviation as a part of the accuracy of the averages.
Sample Calculations (final Lab)	2	Wrote out the calculations I performed in excel, and then also re-adjusted my excel calculations based on my math done by hand.
Power Point Presentation	1	Added the apparatus, and the materials and supplies.
Total	30	

NAME: Basil Hashim

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Prepared gel solution and collected data throughout experiment
Pre-Lab Editing		
Experimental Objective (Prelab)	1	Explained lab goal in the prelab
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	6	Read relevant literature for and wrote discussion of results for the post-lab
References (final and/or prelab)	0.5	Collecting and citing sources used in discussion section of the post-lab
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	0.5	
Total	13	

NAME: Lindsey Meisch

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	Helped prepare the CARBOPOL solution and wrote out procedure
Pre-Lab Editing	2	Edited Procedure and wrote out safety for prelab
Experimental Objective	0	

(Prelab)		
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Wrote out procedure
Project Safety Evaluation (prelab)	1	Analyzed safety precautions for the procedure
Final Lab Editing	1	Edited for cohesion and clearness
Executive Summary (final lab)	0	
Introduction (Final lab)	1	Used lab materials to write out introduction
Theory (Final Lab)	3	Used supplemental lab content to write the theory
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	1	Cited appropriate references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Used the final report to create concise slides to bullet point the major points
Total	18	